

Computational Viability Certification under Incomplete Observation: A Continuous-to-Finite Bridge with Exact Certificates, a Solver-Assisted Certified Campaign, and a Finite-Side Reference Library

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September 24, 2026

Abstract

Nonviability certificates are useful computationally only if they are finite, attainable by standard solvers, and independently checkable. This paper develops such a pipeline for linear systems under exogenous partial observation of a hidden mode. An outer moment approximation of all measurable information-adapted controls, combined with an inner adversarial approximation (finitely stored admissible labels and scenarios), yields a finite linear program whose optimum is a certified lower bound on the continuous-time safety value and whose error inflation is a certified upper bound ($\rho_{\mathcal{H},G} \leq J_{\mathcal{I}}(T) \leq \rho_{\mathcal{H},G} + R_U V_U h_u + \delta_\beta + Lh_t$); every dual-feasible solution yields a finite lower-bound certificate covering every measurable policy, and a strictly positive certified margin is a finite obstruction certificate; along every admissible refinement sequence whose mesh, error, and enclosure widths vanish, nonviability is eventually detected, and whenever a positive certificate exists one can be chosen with at most $r + 1$ safety labels at rank r — the relevant dimension being information–time rank, not instantaneous input dimension, and a matching construction with scalar input produces obstructions whose minimal certificates require arbitrarily many witnesses, saturating the $r + 1$ bound. The certificates account for post-observation recourse: on the three-branch delayed-observation instance every pair of compatible states is viable (explicit pairwise policies are carried exactly) and a common control keeps every branch safe throughout the blind window, yet nonviability is certified, with the exact delay threshold $\tau^* = 7/50$. The finite program is solved by a standard solver (HiGHS) across a five-point mesh refinement; optimality is certified in exact rational arithmetic by independent primal and dual witnesses, the solver agreeing with the exact values to at most 4.2×10^{-17} , and the refinement obeys the closed law $\rho(h) = \max\{0, 3/50 - Th/4\}$. A finite-side reference library, `viacert`, packages the exact finite-horizon recursion and the fibre-partition criteria behind a pluggable finite-system interface, with twelve audited identities in its self-test; the bridge’s continuous programs live in the campaign scripts, not in the library. All exact verification paths for the reported rational benchmark run in integer and rational arithmetic; solver outputs are numerical confirmations, never proof objects.

Keywords: viability; incomplete observation; obstruc-

tion certificates; linear programming; semi-infinite programming; exact arithmetic; certified computation

1. Introduction

Viability theory under full observation has a mature numerical apparatus: kernel approximations, set-valued meshes, and convergence results (Aubin, 1991; Cardaliaguet, Quincampoix, and Saint-Pierre, 1999). Under incomplete observation the sufficiency side is carried by output-feedback conditions and the estimation-tube program (Veliov, 1993), while the necessity side — certificates whose firing *proves* nonviability of an information state (Abaee, 2026b) — has lacked a computational bridge. The difficulty is the quantifier order: the adversary chooses the mode, the initial state inside its cell, and a measurable disturbance with knowledge of the announced policy, so naive discretization of controls proves at most the infeasibility of a held-control subclass, not of the policy class itself.

This paper builds the bridge and the tooling around it. The main result (Theorem 4) approximates the admissible controls from outside by their block moments — a necessary relaxation of continuous-time safety that restricts nothing — and the adversary from inside by explicitly stored scenarios, yielding a finite linear program whose optimal value sandwiches the exact minimax safety violation. Dual solutions of the program are finite, policy-independent obstruction certificates; positive certificates survive mesh refinement; and their size is governed by the rank of the safety-row matrix in the information–time product, which no bound on the instantaneous input dimension controls (Proposition 5).

The pipeline is demonstrated end to end on a three-branch delayed-observation instance: delay threshold, Farkas certificate, and a five-point mesh campaign in which the HiGHS solver (Huangfu and Hall, 2018) attains the exact rational optimum at every mesh, with independent primal–dual witnesses closing the duality gap exactly. The finite-side machinery — the exact backward recursion and the fibre-partition criteria — is packaged as the reference library `viacert` with a pluggable finite-system interface and a self-test of twelve audited identities; the library supplies the enumerate-and-check layer of the certification workflow, and it is deliberately not a reimplementa-

the bridge's continuous programs (Section 10.).

The contributions are: the continuous-to-finite sandwich theorem with sound, refinement-complete, sparsity-bounded certificates (Section 5.); the recourse-aware obstruction reading and the exact three-branch instance with its certified campaign (Section 6.); the information-time rank separation (Section 8.); the refinement-erosion identity and the oracle-recourse extension (Section 9.); the complexity and residual-bounded verification framework (Section 7.); and the reference library (Section 10.).

2. Model, information, and quantifiers

Dynamics. Let the hidden mode be $\theta \in \Theta := \{1, \dots, M\}$. Conditional on θ , the continuous state satisfies

$$\dot{x}(t) = A_\theta(t)x(t) + g_\theta(t) + B_\theta(t)u(t) + E_\theta(t)d(t), \quad t \in [0, T],$$

under the standing assumptions: $A_\theta, g_\theta, B_\theta, E_\theta$ bounded and piecewise continuous, with bounded-variation bounds on B_θ, E_θ — the operative assumption below is the induced bound $\text{TV}(k_a) \leq V_U$ on the adjoint kernels, assumed directly in Theorem 4 rather than derived from the primitive coefficients here; the initial set in mode θ a nonempty compact polytope $P_\theta \subset \mathbb{R}^n$; the input and disturbance sets nonempty compact polytopes $U = \{v : Fv \leq f\} \subset \mathbb{R}^m$ and $D_\theta \subset \mathbb{R}^p$; safety the closed polyhedron $\mathcal{V} = \{x : c_i^\top x \leq b_i, i = 1, \dots, s\}$; and the initial information state $B_0 = \bigcup_\theta \{\theta\} \times P_\theta$ initially safe, meaning $P_\theta \subseteq \mathcal{V}$ for every θ (if some P_θ leaves $\mathcal{V}$ at time zero, a time-zero certificate suffices and the analysis is vacuous). The dynamics admit unique Carathéodory solutions for every admissible measurable signal pair.

Information. At known times $0 = \tau_0 < \tau_1 < \dots < \tau_R = T$ the controller receives exogenous information about θ : refining history partitions $\mathcal{P}_0 \preceq \mathcal{P}_1 \preceq \dots \preceq \mathcal{P}_{R-1}$ of Θ — the order is by refinement: for $r' > r$ every cell of $\mathcal{P}_{r'}$ lies inside a cell of $\mathcal{P}_r$ (the last partition is $\mathcal{P}_{R-1}$ because $\tau_R = T$ ends the last interval) — the observation history during $I_r = [\tau_r, \tau_{r+1})$ identifying the cell $C_r(\theta) \in \mathcal{P}_r$. These are history partitions, not necessarily partitions generated by the latest observation alone. Observations do not depend on x, u, d ; active sensing and endogenous observation branching are excluded from the exact theorem (Section 9. gives a sound extension).

Policies and quantifiers. Policies are deterministic, measurable, nonanticipative, and adapted to the observation history; every such policy is represented exactly by signals $u_{r,C} \in L^\infty(I_r; U)$, $C \in \mathcal{P}_r$. The robust quantifier order is

$$\exists \pi \in \Pi_{\mathcal{I}} \forall \theta \forall x_0 \in P_\theta \forall d \in L^\infty([0, T]; D_\theta) \forall t \in [0, T],$$

the adversary selecting mode, initial state, and measurable disturbance with knowledge of the announced policy; randomized policies are not included. Throughout, a belief is a nonempty set $B \subseteq \Theta \times \mathbb{R}^n$ of mode-state possibilities, not a probability distribution. The information-constrained viability kernel is $\mathcal{K}_{\mathcal{I}}^T = \{B : \exists \pi \forall \theta, x_0, d, t \quad x_{\theta, x_0, d}^\pi(t) \in \mathcal{V}\}$, a set of beliefs; the audited initial state B_0 is viable exactly when $B_0 \in \mathcal{K}_{\mathcal{I}}^T$. Writing $\text{Lift}(K_{\text{full}}^T) = \{B \subseteq \Theta \times \mathbb{R}^n : \emptyset \neq B \subseteq K_{\text{full}}^T\}$ for the belief lifting of the full-information kernel (an ambient $\Theta \times \mathbb{R}^n$ -valued family), $\mathcal{K}_{\mathcal{I}}^T \subseteq \text{Lift}(K_{\text{full}}^T)$, and the inclusion is strict in general: the instance of Section 6. has every two-state subbelief viable — with the pairwise policies computed exactly there — while the three-state belief is not.

3. Adjoint safety rows and the safety value

Let $X_\theta(t, s)$ denote the state-transition matrix. For a label $a = (\theta, i, t) \in \mathcal{S} := \Theta \times \{1, \dots, s\} \times [0, T]$ define the input kernel and margin

$$k_a(\sigma) = \mathbf{1}_{\{\sigma \leq t\}} B_\theta(\sigma)^\top X_\theta(t, \sigma)^\top c_i,$$

$$\begin{aligned} \beta_a &= b_i - h_{P_\theta}(X_\theta(t, 0)^\top c_i) - \int_0^t c_i^\top X_\theta(t, \sigma) g_\theta(\sigma) d\sigma \\ &\quad - \int_0^t h_{D_\theta}(E_\theta(\sigma)^\top X_\theta(t, \sigma)^\top c_i) d\sigma, \end{aligned}$$

where $h_Q(v) = \sup_{q \in Q} v^\top q$. For a policy π set $\mathcal{F}_a(\pi) = \int_0^T k_a(\sigma)^\top u_\theta^\pi(\sigma) d\sigma - \beta_a$.

Proposition 1 (robust rows). *π is robustly safe on $[0, T]$ if and only if $\mathcal{F}_a(\pi) \leq 0$ for every $a \in \mathcal{S}$. This follows from the linear solution formula; initial-state and disturbance maximizations separate from the control term because the control signal in a mode does not depend on x_0 or d . For polytopic D_θ , a maximizing disturbance can be chosen measurably by selecting a maximizing vertex at each time, ties resolved by a fixed ordering.*

Define the minimax safety violation $J_{\mathcal{I}}(T) = \min_{\pi \in \Pi_{\mathcal{I}}} \max_{a \in \mathcal{S}} \mathcal{F}_a(\pi)$.

Proposition 2 (value and nonviability). *The minimum exists: the policy-signal space is a finite product of weak-* compact sets $L^\infty(I_r; U)$ (each U a compact convex polytope, so the admissible signal sets are weak-* closed and bounded) and each $\mathcal{F}_a$ is weak-* continuous; the maximum over labels is attained on the compact label set $\mathcal{S}$ because $a \mapsto \mathcal{F}_a(\pi)$ is continuous in t under assumption (6) below. Consequently $B_0 \notin \mathcal{K}_{\mathcal{I}}^T$ if and only if $J_{\mathcal{I}}(T) > 0$. The strict positive margin is a consequence of compactness and closed safety constraints, not an extra nonviability assumption.*

4. The finite obstruction certificate

Define $\phi_U(g) := \min_{v \in U} g^\top v = -h_U(-g)$. Choose finitely many labels $a_\ell = (\theta_\ell, i_\ell, t_\ell)$, $\ell = 1, \dots, q$, and weights $\lambda_\ell \geq 0$, $\sum_\ell \lambda_\ell = 1$. For each active information cell $C \in \mathcal{P}_r$, pool the adjoint kernels $g_{r,C}(\sigma) = \sum_{\ell: \theta_\ell \in C} \lambda_\ell k_{a_\ell}(\sigma)$. The certificate margin is

$$\Gamma_{\mathcal{I}}(\lambda, a) = \sum_{r=0}^{R-1} \int_{I_r} \sum_{C \in \mathcal{P}_r} \phi_U(g_{r,C}(\sigma)) d\sigma - \sum_{\ell=1}^q \lambda_\ell \beta_{a_\ell}. \quad (1)$$

Remark 3 (semantics). If $\Gamma_{\mathcal{I}}(\lambda, a) > 0$, then no admissible information-adapted policy is safe: for every policy, $\sum_\ell \lambda_\ell \mathcal{F}_{a_\ell}(\pi) \geq \Gamma_{\mathcal{I}}(\lambda, a)$, so at least one stored label satisfies $\mathcal{F}_{a_\ell}(\pi) \geq \Gamma_{\mathcal{I}}(\lambda, a) > 0$ — an admissible realization violates facet i_ℓ at time t_ℓ . The violating label may depend on the policy; the finite list and its weights do not. The certificate is not an adversarial tree: it is a finite, policy-independent set of adversarial tests combined by an adjoint identity. The pooled bound is tight as a minimum over policies: $\min_\pi \sum_\ell \lambda_\ell \mathcal{F}_{a_\ell}(\pi) = \Gamma_{\mathcal{I}}(\lambda, a)$, the pointwise minimizer being measurable by fixed-order vertex selection in the polytope U .

5. The continuous-to-finite theorem

The exact dual expression (1) alone would be close to standard infinite-dimensional convex duality (Anderson and Nash, 1987; Hettich and Kortanek, 1993). The consequential result is a sound finite construction covering unrestricted measurable controls.

Outer control moments. Choose a control-time mesh $\mathcal{H} = \{J_1, \dots, J_H\}$ containing every sensing time. Introduce one vector $v_{j,C} \in U$ for every cell $C \in \mathcal{P}_{r(j)}$ active during J_j , where $r(j)$ is the unique observation interval containing J_j (the mesh contains every sensing time, so each block lies in one I_r), and let $Q = \sum_j |\mathcal{P}_{r(j)}|$ be the number of information-time blocks. For a label a define the finite row $A_{a,j,C} = \mathbf{1}_{\{\theta \in C\}} \int_{J_j} k_a(\sigma) d\sigma$, the block average $\bar{k}_{a,\mathcal{H}}(\sigma) = \frac{1}{|J_j|} \int_{J_j} k_a(s) ds$ on J_j , and the moment error

$$e_{a,\mathcal{H}} = \int_0^T h_U(\bar{k}_{a,\mathcal{H}}(\sigma) - k_a(\sigma)) d\sigma, \quad (2)$$

for which any verified upper bound $\bar{e}_a \geq e_{a,\mathcal{H}}$ may be used. For every measurable policy, its block averages belong to U , and

$$\sum_{j,C} A_{a,j,C}^\top v_{j,C} - \int_0^T k_a(\sigma)^\top u_{\theta(a)}^\pi(\sigma) d\sigma \leq \bar{e}_a. \quad (3)$$

Inequality (3) is the crucial orientation: finite feasibility is a necessary condition for continuous-time safety. The moments do not restrict the real controller to held inputs. For the sampled-data subclass whose policies are held constant on the blocks, the same polyhedron is an exact reduction rather than a relaxation.

Inner adversarial approximation. One may use a right-hand side $\beta_a^+ \geq \beta_a$ obtained from an explicit admissible initial state and disturbance signal rather than exact worst-case optimization. If $0 \leq \beta_a^+ - \beta_a \leq \delta_\beta$ uniformly over the labels used, the approximation is safe for nonviability: it weakens the adversary and enlarges the feasible controller set. For example, with $w_a(\sigma) = \mathbf{1}_{\{\sigma \leq t\}} E_\theta(\sigma)^\top X_\theta(t, \sigma)^\top c_i$, on a disturbance mesh choose $d_{a,J} \in \arg \max_{d \in D_\theta} (\int_J w_a)^\top d$, yielding an explicit piecewise-constant admissible disturbance; if D_θ has radius R_D about a fixed center and $\text{TV}(w_a) \leq V_D$, then

$$0 \leq \beta_a^+ - \beta_a \leq R_D V_D h_d, \quad (4)$$

the general budget being $\delta_\beta = \delta_{\text{init}} + \delta_{\text{dist}} + \delta_{\text{num}}$ for separately accounted initial-state and numerical error.

The finite program. Let $G \subset [0, T]$ be a finite time grid with $0, T \in G$ and every observation time in G ; the label set is the product $\mathcal{S}_G = \Theta \times \{1, \dots, s\} \times G$, with covering radius $h_t = \sup_{t \in [0, T]} \min_{\hat{t} \in G} |t - \hat{t}|$. Solve

$$\rho_{\mathcal{H},G} = \min_{v,\rho} \rho \quad \text{s.t.} \quad A_a v - \beta_a^+ - \bar{e}_a \leq \rho \quad (a \in \mathcal{S}_G), v_{j,C} \in U. \quad (5)$$

Theorem 4 (continuous-to-finite). *Assume: uniformly over admissible policies, modes, and facets, $|\mathcal{F}_{\theta,i,t}(\pi) - \mathcal{F}_{\theta,i,t'}(\pi)| \leq L|t - t'|$ (6); the input kernels satisfy $\text{TV}(k_a) \leq V_U$ (7); U has radius R_U about some fixed center; and the time grid has covering radius at most h_t , the control mesh maximum length h_u , and the adversarial approximation satisfies the bound above. Then:*

(i) *Certified value sandwich.*

$$\rho_{\mathcal{H},G} \leq J_{\mathcal{I}}(T) \leq \rho_{\mathcal{H},G} + R_U V_U h_u + \delta_\beta + L h_t. \quad (8)$$

With (4), $J_{\mathcal{I}}(T) - \rho_{\mathcal{H},G} \leq R_U V_U h_u + R_D V_D h_d + L h_t$. Verified numerical row widths are added to the right side.

(ii) *Sound finite certificates.* If $\rho_{\mathcal{H},G} > 0$, the program's dual supplies a finite certificate of nonviability of the original continuous-time system, covering every measurable information-adapted policy.

(iii) *Completeness under refinement.* If $B_0 \notin \mathcal{K}_{\mathcal{I}}^T$, then for every admissible refinement sequence along which h_u, h_d, h_t and all numerical enclosure widths converge to zero, the meshes eventually produce a positive finite certificate. Nonviability is thus semidecidable in this class for the input model in which the coefficients are exact rational or interval-enclosed piecewise-polynomial data with certified integration, total-variation, and uniform time-Lipschitz bounds: the semidecision procedure enumerates admissible meshes, solves each enclosed rational program, and halts on the first positive certificate.

(iv) *Exact atomic dual representation.*

$$J_{\mathcal{I}}(T) = \sup_{q < \infty, a_1, \dots, a_q, \lambda \in \Delta_q} \Gamma_{\mathcal{I}}(\lambda, a). \quad (9)$$

In particular, nonviability is equivalent to the existence of a positive finite atomic certificate.

(v) *Sparse witnesses.* Let $r = \text{rank } A$ for the finite safety-row coefficient matrix of (5) — the A_a rows excluding the ρ -column and the actuator inequalities; the bound is LP basic-solution theory (supplementary Step 6). Whenever $\rho_{\mathcal{H},G} > 0$, a positive certificate exists using at most $r + 1 \leq mQ + 1$ safety labels, and it can retain any prescribed margin strictly below $\rho_{\mathcal{H},G}$. If the lifted continuous-time kernels span a finite-dimensional space of dimension $r_{\mathcal{I}}$, a continuous-time obstruction has at most $r_{\mathcal{I}} + 1$ labels. The relevant dimension is information-time rank, not m .

Proof outline. The lower bound holds because every admissible measurable policy's control bounds are outer-approximated by the block moments, so the moment program relaxes the adversary's feasible set and its value cannot exceed the continuous safety value; the upper bound is realized by taking, on each block, the piecewise-constant control attained by the finite optimizer's block variables, whose exit from the continuous value is charged exactly by the three error terms — the control-moment relaxation $R_U V_U h_u$, the stored-scenario error δ_β , and the time-grid interpolation error $L h_t$. The asymmetry at its core is worth stating plainly: averaging arbitrary controls proves the lower bound; implementing piecewise-constant controls proves the upper bound. Confusing these two directions is what makes naïve infeasibility-by-discretization arguments unsound.

6. A three-branch delayed-observation instance carried exactly

Plant and exact delay threshold. Consider the planar double integrator $\dot{p} = v$, $\dot{v} = u$, with unit vectors $n_1 = (1, 0)$, $n_2 = (-\frac{3}{5}, \frac{4}{5})$, $n_3 = (-\frac{3}{5}, -\frac{4}{5})$, the hexagonal input set $U = \text{conv}\{\pm n_1, \pm n_2, \pm n_3\}$, equivalently $\{v : |u_2| \leq \frac{4}{5}, |2u_1 + u_2| \leq 2, |2u_1 - u_2| \leq 2\}$, safety $n_i^\top p \leq 2$ ($i = 1, 2, 3$) and $|v_1|, |v_2| \leq \frac{6}{5}$ — seven facets in

all, $s = 7$ (three position, four velocity) — three possible initial states $p_j(0) = \frac{34}{25}n_j$, $v_j(0) = n_j$, no disturbance, and the mode unobserved until time τ , when it is revealed exactly.

If j is known, $u = -n_j$ for one second and then $u = 0$ keeps the maximum critical position at $\frac{34}{25} + \frac{1}{2} = \frac{93}{50} = 1.86 < 2$: all three physical states are indefinitely viable under full information. Take three witness labels, each using its own position facet at $t_* = \tau + 1$, with kernels $k_j(s) = \mathbf{1}_{\{s \leq t_*\}}(t_* - s)n_j$ and $\beta_j = \frac{16}{25} - t_*$, weighted by $\lambda_1 = \frac{3}{8}$, $\lambda_2 = \lambda_3 = \frac{5}{16}$, which satisfy $\sum_j \lambda_j = 1$ and $\sum_j \lambda_j n_j = 0$. Before observation the pooled kernel is identically zero; after observation each mode has its own control and $\phi_U(\lambda_j(t_* - s)n_j) = -\lambda_j(t_* - s)$. Hence

$$\Gamma(\tau) = - \int_{\tau}^{\tau+1} (\tau+1-s) ds - \left(\frac{16}{25} - \tau - 1 \right) = \tau - \frac{7}{50}, \quad (10)$$

so $\tau > \frac{7}{50}$ implies $B_0 \notin \mathcal{K}_{\mathcal{I}}$. Conversely, $u = 0$ before observation and $u = -n_j$ for one second afterward keeps the maximum critical position at $\frac{34}{25} + \tau + \frac{1}{2}$, safe exactly when $\tau \leq 0.14$. Thus $\tau_{\max} = \frac{7}{50} = 0.14$ is exact (equality viable: the safety set is closed), and for horizons $T \geq \tau + 1$ the upper-bound policy — full braking from τ on, stopping at $\tau + 1$ from $p = (\frac{34}{25} + \tau)n_j$ — shows every row value at most $\tau - \frac{7}{50}$, so $J_{\mathcal{I}}(T) = \tau - \frac{7}{50}$.

Pairs are viable; the blind-window exit test misses. At $\tau = \frac{1}{5}$, the common control $u = 0$ keeps all three branches safe throughout the blind window: $n_j^{\top} p_j(0.2) = \frac{39}{25} = 1.56 < 2$. The pair claims are carried exactly. For $\{1, 2\}$ (and $\{1, 3\}$ by the reflection $v_2 \mapsto -v_2$): hold $u = 0$ through the blind window, then brake each branch along its own normal — $u = -\frac{2}{5}n_j$ on $[\tau, \tau + \frac{1}{5}]$, then $u = -n_j$ to rest — so the critical position peaks at most at $\frac{12345}{6250} = 1.9752 < 2$. For $\{2, 3\}$: hold $u = n_1$ through the blind window, then brake each branch along its own normal — $u = -\frac{3}{5}n_j$ on $[\tau, \tau + \frac{1}{5}]$, then $u = -n_j$ to rest — so the critical position peaks at most at $\frac{2419}{1250} = 1.9352$. All seven facets hold at every time on every pair-policy trajectory (the velocity boundary $|v_1| = \frac{6}{5}$ is met exactly by branch 1 before τ , and the constraints are closed). No exit occurs before information arrives; nevertheless $\Gamma(0.2) = 0.06 > 0$ — the best post-observation recourse cannot recover all branches. The obstruction is not that information arrives after unavoidable physical exit; it is that information arrives after some indistinguishable branch has necessarily accumulated more braking liability than the remaining control authority can discharge.

The finite program and its exact witnesses. Set $\tau = \frac{1}{5}$, $T = \frac{6}{5}$, $h = \frac{1}{10}$: two common pre-observation control blocks and ten post-observation blocks for each of three modes, so $Q = 32$, $mQ = 64$. For the endpoint position rows the exact moment error is $e = 12 \cdot h^2/4 = \frac{3}{100}$ (the kernel is affine on each of the $T/h = 12$ mesh intervals and $h_U(\alpha n_j) = |\alpha|$); the endpoint right-hand side is $\beta = \frac{16}{25} - \frac{6}{5} = -\frac{14}{25}$, relaxed to $\beta + e = -\frac{53}{100}$. The exact continuous certificate has margin $\frac{3}{50} = 0.06$; the finite moment certificate has margin $\frac{3}{100} = 0.03$.

The dual witness is exact. Order the input facets by $F = [(0, 1); (0, -1); (2, 1); (2, -1); (-2, 1); (-2, -1)]$, $f = (\frac{4}{5}, \frac{4}{5}, 2, 2, 2, 2)$, and define $\eta_1 = (0, 0, 0, 0, \frac{1}{4}, \frac{1}{4})$, $\eta_2 = (0, \frac{1}{2}, 0, \frac{3}{10}, 0, 0)$, $\eta_3 = (\frac{1}{2}, 0, \frac{3}{10}, 0, 0, 0)$, satisfying $F^{\top} \eta_j = -n_j$ and $f^{\top} \eta_j = 1$. For a post-observation interval J_k put $a_k = \int_{J_k} (T - s) ds$ and $\mu_{j,k} = \lambda_j a_k \eta_j$. Before observation,

stationarity holds because $\sum_j \lambda_j n_j = 0$; after observation, because $\lambda_j a_k n_j + F^{\top} \mu_{j,k} = 0$. Since $\sum_k a_k = \frac{1}{2}$, the Farkas contradiction is $\lambda^{\top}(\beta + e) + \mu^{\top} f_{\text{blk}} = -\frac{53}{100} + \frac{1}{2} = -\frac{3}{100} < 0$ — an exact rational certificate. The primal witness is equally exact: hold the pre-observation blocks at 0 and the post-observation blocks of mode j at $-n_j$; the endpoint row then reads $-\frac{1}{2} + \frac{14}{25} - \frac{3}{100} = \frac{3}{100}$, and every other row lies at or below the same value; the campaign script checks all 465 rows exactly at this mesh, and the row slacks are archived with it (Section 13.).

Solver-assisted, exactly certified mesh study. For each mesh the program (5) is built at its claimed dimensions and solved by HiGHS; optimality is then certified in exact rational arithmetic by the witness pair above (primal feasibility of every row; dual stationarity and dual objective $-\rho(h)$), so the duality gap closes exactly and the solver's returned value is a computational confirmation of a proved optimum, not an observation. Table 1 reports the campaign: exact values $\rho(h) = \max\{0, \frac{3}{50} - Th/4\}$, solver values, iteration counts, and wall-clock times as measured on the verification machine.

On the audited family of uniform meshes $h \in \{\frac{1}{5}, \frac{1}{10}, \frac{1}{20}, \frac{1}{50}, \frac{1}{100}\}$ (each dividing both τ and T), the values obey the exact law $\rho(h) = \frac{3}{50} - Th/4$ for $h < \frac{1}{5}$ and vanish at $h = \frac{1}{5}$: the whole refinement gap is the control-moment error $Th/4$ of the endpoint rows, and the floor at zero says the finite value cannot drop below the exact continuous margin once the moment error vanishes. All dimension counts ($mQ + 1$ variables; $Ms(N_t + 1) + p_U Q$ inequalities), all witness inequalities, and the closed-form law are machine-verified in exact rational arithmetic by the verification scripts of Section 13., which regenerate Table 1 verbatim and re-run the solver campaign.

7. Computation, complexity, and independent verification

With Q information-time blocks and $N_t + 1$ time labels, the program has $mQ + 1$ variables and at most $Ms(N_t + 1) + p_U Q$ inequalities (p_U input-polytope facets); it is polynomial-time in these dimensions and the encoded rational data size (weakly polynomial bit complexity) for the exact rational benchmark programs, and for interval-enclosed inputs once the enclosures are fixed — conditional on the validated coefficient bounds and certified row integrals. Initial-state dependence is affine, so no vertex enumeration of P_{θ} is required: at each label solve $\max_{\xi \in P_{\theta}} c_i^{\top} X_{\theta}(t, 0)\xi$ and store a maximizing or near-maximizing point — the verifier needs only membership of the stored point in P_{θ} , not global optimality; near-optimality bears on the convergence guarantee, not on a positive certificate's validity. The same holds for stored disturbance signals.

For rational outer rows $\hat{A}v \leq \hat{b}$, $F_{\text{blk}}v \leq f_{\text{blk}}$, a residual-bounded certificate test suffices: with $\lambda, \mu \geq 0$, $r = \hat{A}^{\top} \lambda + F_{\text{blk}}^{\top} \mu$, $c = \lambda^{\top} \hat{b} + \mu^{\top} f_{\text{blk}}$, the condition

$$c + \sum_j h_U(-r_j) < 0 \quad (11)$$

is sufficient, since any feasible v satisfies $-\sum_j h_U(-r_j) \leq r^{\top} v \leq c$: the upper bound is the multiplier combination, the lower bound is termwise minimization over U , and the

h	blocks Q	variables $mQ+1$	rows $Ms(N_t+1) + p_U Q$	exact $\rho(h)$	solver ρ	iterations	time (s)
0.20	16	33	243	0	0.000000000	36	0.024
0.10	32	65	465	$\frac{3}{190} = 0.030$	0.030000000	78	0.014
0.05	64	129	909	$\frac{200}{200} = 0.045$	0.045000000	141	0.043
0.02	160	321	2241	$\frac{27}{590} = 0.054$	0.054000000	501	0.252
0.01	320	641	4461	$\frac{57}{1000} = 0.057$	0.057000000	1062	1.015

Table 1: Solver-assisted, exactly certified mesh study on the three-branch instance ($\tau = \frac{1}{5}$, $T = \frac{6}{5}$): program dimensions, exact optimal values, HiGHS-attained values (agreement to at most 4.2×10^{-17}), simplex iterations, and wall-clock time (single runs, scipy HiGHS dual simplex, on the verification machine; iteration counts and times may vary with solver version and settings; “rows” counts LP inequalities — safety plus actuator, with $s = 7$, $p_U = 6$, $N_t = T/h$). Optimality of each exact value is certified by the exact primal–dual witness pair, independent of the solver. At $h = 0.2$ the witness is an exact optimality witness of value zero; at the four finer meshes it is a positive obstruction certificate. The exact continuous value is 0.06; the three-label support of each positive certificate is an instance-specific fact of this symmetric campaign, distinct from the general $r + 1$ bound.

two displayed inequalities make the sign convention auditable. The actuator rows $F_{\text{blk}} v \leq f_{\text{blk}}$ already represent $v_j \in U$; their multipliers in the residual test are redundant but legitimate, and the verifier uses both paths as independent cross-checks. All integration, rounding, and model-enclosure errors must be included before applying the test, and uncertain row data must be relaxed toward a larger feasible controller set: overestimating the worst disturbance contribution can create a false obstruction — the sign convention of the $\beta_a^+ \geq \beta_a$ relaxation of Section 5.: a nonmaximizing stored disturbance gives a larger β_a^+ , hence a weaker constraint, which is the sound direction for a lower bound on $J_{\mathcal{I}}(T)$. A deletion-based sparsification attempts supported labels for deletion one at a time in a fixed order, keeping a deletion only if the prescribed target margin survives the feasibility re-solve; it terminates with an irredundant (not claimed minimal) supported label set, at one feasibility solve per safety row; exact rank compression requires verified rank, and floating-point singular-value truncation must be charged to the error budget. If $J_{\mathcal{I}}(T) = \gamma > 0$, a mesh with $R_U V_U h_u + R_D V_D h_d + L h_t < \gamma$ guarantees detection — an explicit margin-dependent resolution bound (mesh resolution as a function of the margin γ), not a certificate-size bound in general.

8. Information–time rank: input dimension does not bound the witnesses

Proposition 5 (arbitrarily large minimal obstructions with one input). *For every integer $q \geq 1$ there is a continuous-time system with scalar input $u(t) \in [0, 1]$ and $q + 1$ indistinguishable modes such that every proper sub-belief is viable, the full $(q+1)$ -mode belief is not, and every obstruction certificate must involve all $q + 1$ modes.*

Proof. Let $T = q$, $I_j = [j - 1, j]$. Mode $j \leq q$ carries the scalar state x obeying $\dot{x} = \mathbf{1}_{I_j}(t)(u - 1)$, $x(0) = 0$; mode $q + 1$ carries $\dot{x} = -u$, $x(0) = q - \varepsilon$, $0 < \varepsilon < 1$. Require $0 \leq x \leq q$ — the terminal bounds are the binding ones, each x being monotone along its mode — and there are no observations. Writing $w_j = \int_{I_j} u$, safety requires $-w_j \leq -1$ for $j \leq q$ and $\sum_j w_j \leq q - \varepsilon$; summing gives $0 \leq -\varepsilon$, a contradiction. If mode $q + 1$ is removed, $u \equiv 1$ is safe; if mode $j \leq q$ is removed, $u = 0$ on I_j and $u = 1$ elsewhere is safe. A certificate supported on fewer than $q + 1$ modes would also certify the corresponding proper subbelief, contradicting its viability, so every obstruction certificate involves all $q + 1$ modes. Equal weights give

certificate margin $\varepsilon/(q + 1)$; the kernel rank is q , and $q + 1$ witnesses are necessary. $\square$

A statement such as “at most $m + 1$ compatible states suffice to witness nonviability” is valid for convex common-action sets in $\mathbb{R}^m$ but false for a common blind control function: the correct dimension counts independent temporal and informational control decisions. This distinction is consequential for review intervals, delayed sensing, and certificate-size claims; its discrete counterpart is a scoping caveat of the finite audits of Abaee (2026a).

9. Refinement erosion and oracle recourse

Suppose a coarse information cell C is split into finer cells $C_1, \dots, C_k$. Because $\phi_U(g_1 + \dots + g_k) \geq \sum_j \phi_U(g_j)$, the same witness satisfies $\Gamma_{\text{coarse}} \geq \Gamma_{\text{fine}}$, with the exact difference

$$\int \left[\phi_U \left(\sum_j g_{C_j} \right) - \sum_j \phi_U(g_{C_j}) \right] dt. \quad (12)$$

For $U = [-\bar{u}, \bar{u}]^m$ this is $\bar{u} \int [\sum_j \|g_{C_j}(t)\|_1 - \|\sum_j g_{C_j}(t)\|_1] dt$: the certificate identifies the precise time intervals on which indistinguishable branches demand conflicting control effort. Equation (12) is the change in the same witness’s margin under refinement, not a claim about the optimized viability value: a coarse cell can strengthen an obstruction, and splitting it can dissolve the certificate while the system’s verdict is unchanged. The inequality is proved above by pointwise minimization; on rational instances it is additionally exactly checkable, and the verification record confirms $\phi_U(g_1 + g_2) \geq \phi_U(g_1) + \phi_U(g_2)$ on 3,240 deterministic rational pairs for the hexagonal U of Section 6. as a software regression test, not as evidence for the identity.

The exact theorem assumes exogenous mode observations. A useful sound extension does not: if finitely many admissible initial-state/disturbance scenarios are provably indistinguishable until τ , give the controller after τ separate arbitrary control signals for every stored scenario — an oracle relaxation of the true policy class. Scenarios are stored initial-state/disturbance pairs, indistinguishable through τ in the sense that they generate identical observation histories; after τ the oracle reveals the realized scenario, and separate controls are granted even where scenarios remain observationally identical. The soundness direction: every real output-feedback policy induces one common pre- τ control, the oracle class contains every

such policy, and nonviability in the enlarged class transfers down. If

$$\int_0^\tau \phi_U\left(\sum_\ell \lambda_\ell k_\ell\right) dt + \sum_\ell \int_\tau^T \phi_U(\lambda_\ell k_\ell) dt - \sum_\ell \lambda_\ell \beta_\ell > 0, \quad (13)$$

with β_ℓ computed from the explicitly stored scenario, then no actual output-feedback policy can be safe. This covers arbitrary sensing after a known blackout; it is sound but generally incomplete.

10. The reference library

The finite side of the pipeline is packaged as **viacert**, a standard-library-only Python package (no third-party dependencies) using exact integer and rational arithmetic. Its design follows three principles. *Exactness*: the finite audits and exact certificate verifiers use integer and rational arithmetic; floating-point solver output is never used as a proof object. *Fidelity to the verified recursion*: the finite-horizon selector is the calculus’s backward recursion (Abaee, 2026b) with the no-repeat protocol threaded as a parameter, matching the audited implementations rather than a reformulation. *Pluggability*: new systems enter through a **FiniteSystem** interface (states, actions, safe set, transition, null action); the audited systems — the shared two-coordinate audit system, the continuous benchmark caps, the certainty-equivalence law pair, and the review grid — ship as default instances.

The public surface comprises the recursion module (viability verdicts, kernels, and step-one witness extraction), the monitoring module (safe and common safe-action sets, the partition-form adequacy converse, maximal common-action sets, the partition census, the review-timing identity with its threshold convention, and the certainty-equivalence drift audit), and a command-line self-test of twelve audited identities — the four-cell kernel counts 24/26/25/28 with the meet identity, the census 15 total and 7 adequate partitions with maximal sets overlapping in (2,2), the benchmark $Y^* = 27/5$, the drift bounds 21/100 and 0, the 48-cell delay identity with its boundary cell, and witness extraction — each a pointer to the published audits (Abaee, 2026a, 2026b). The inherited counts are software self-tests pointing to the published audits; they are not used in this paper’s main theorem or example, and their full context lives in the package documentation. In the certification workflow the library plays the enumerate-and-check role: finite instances are enumerated exactly, candidate designs are tested against the fibre criteria, and every claim is archived with the script that regenerates it. The library is not a reimplement of this paper’s bridge theorems: it carries the exact finite machinery they reduce from, while the bridge’s programs live in the campaign scripts of Section 13; consolidating those scripts as a **viacert.continuous** module is the library’s recorded next step. Consolidation supersedes implementation paths only in future releases: the archived scripts and their audited identities remain the exact versions used for this article’s verification record. The title’s “reference library” is read as this finite-side package plus the certified campaign pipeline.

11. Scope delimitations

The exact theorem does not cover: control-dependent observations or active sensing; arbitrary continuous-state measurements; nonlinear dynamics without additional validated enclosures; nonconvex actuator sets without a relaxation gap (convexity is structural: block averages of admissible controls lie in $\text{conv } U$, and the weak-* compactness of Proposition 2 uses that U is a compact convex polytope); rate-constrained actuators without modifying the approximation; or unrestricted infinite-horizon completeness. For general sensing after a blackout, the oracle-recourse certificate is sound but incomplete — exactness in the audited instance belongs to the complete revelation at τ , not to the relaxation. For nonlinear systems, a uniform validated flow-error bound can be added to the row budgets, yielding a sound extension, not automatic completeness. Against the alternatives: full-information viability is necessary but not sufficient for the information-constrained question; initial-admissibility and tangency tests are silent on the instance’s obstruction, which is temporal-informational rather than pointwise-geometric; safety-before-first-observation tests are silent when a common safe blind control exists; exact finite safety games exist as discrete models, and what is needed to use them as sound abstractions of the present continuous systems is a justified continuous-to-finite bridge, which is what Theorem 4 provides; and naïve held-control infeasibility excludes only the held-control class. No general-purpose computational-calculus claim is made: the theorem is a certificate generator for the stated class, with completeness only under refinement and semidecidability only with effectively computable bounds.

12. Conclusion

A sound and independently checkable certification pipeline under incomplete observation rests on three requirements: the finite controller class must be an outer relaxation while stored adversarial tests correspond to genuinely admissible scenarios; optimality must be certifiable independently of the solver; and certificate size must be understood in the information–time product rather than the input dimension. The bridge theorem delivers all three, the three-branch instance shows the resulting certificates detecting failure that full-information viability, blind-window exit tests, and held-control infeasibility all miss, and the campaign shows the solver returning the independently proved optima to numerical precision while exact witnesses close every gap. The reference library carries the exact finite machinery to new systems. What remains open is completeness beyond refinement: an inner approximation of the policy class that is computationally effective for general sensing is not known, and the oracle-recourse extension is the current sound surrogate.

13. Verification methods

Every reported mathematical benchmark value and every table entry is regenerated by deterministic scripts, exact rational arithmetic unless stated as solver output, standard library plus the solver interface only. Three scripts and one package cover the claims, with one master entry

point chaining them. The exact verification record re-derives the instance’s certificates — delay threshold, margins, witnesses, erosion identity, rank construction — in 51 exact checks. The campaign script rebuilds the program (5) at each of the five meshes, runs HiGHS, and certifies optimality by the exact primal–dual witnesses, closing the duality gap; it re-derives the mesh law and all dimension identities (nine checks). The library self-test runs the twelve audited identities. A master script chains the three, then recomputes independently: the mesh values from the closed law $\max\{0, 3/50 - Th/4\}$, the Farkas contradiction $-\frac{53}{100} + \frac{1}{2} = -\frac{3}{100}$, the primal row value $\frac{3}{100}$, the margins $\frac{3}{50}$ and $\frac{3}{100}$, the moment error $e = \frac{3}{100}$, the relaxed right-hand side $-\frac{53}{100}$, the delay threshold $\tau^* = \frac{7}{50}$, the blind-window safety value $\frac{39}{25}$, and every headline number in the text; independence here means a separate formula path from the subordinate scripts, not a separate implementation — an independent certificate-file parser that reads the archived witnesses without importing the model builder is the recorded next step (Section 10.).

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Declarations

Funding

None.

Competing interests

None.

Data availability

No external data were used.

Code availability

The verification scripts `paper2_computational_certification_v2_verification.py`, `paper2_nonstandard_verification.py`, `paper2c_lp_campaign.py`, and the `viacert` package are publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`, at the commit pinned to this article’s submission); they reproduce all exact values, witnesses, bounds, and the solver campaign verbatim.

AI declaration

GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review. The author reviewed and edited outputs and takes responsibility for the final work.